6-1 Analog and Digital

Point!

Analog and Digital

Analog



Digital



(1) ([1]**Analog**) : Refers to quantities that can be measured continuously and with fine precision, such as mass, time, or temperature. Data represented in analog form is referred to as ([2]**analog data**).

(2) ([3]**Digital**) : Refers to quantities that continuously change and are represented numerically by segmenting at regular intervals. Data represented in digital form is referred to as ([4]**digital data**).

(3) ([5]**A/D conversion (digitization)**) : The process of converting analog data into digital data.

(4) ([6]**D/A conversion**) : Converting digital data into analog data.

(5) Advantages of Digital Data

[1] Data can be ([7]**accumulated**) or ([8]**duplicated**) without degradation.

[2] It is easy to ([9]**modify**) and ([10]**edit**) the data.

[3] It is possible to efficiently ([11]**communicate**) the information.

[4] It is possible to ([12]**combine**) various types of media.

(6) ([13]**Binary**) : A method of representing numbers using two types of digits, 0 and 1.

- In computers, all types of information, such as numbers, characters, audio, and images, are represented in ([14]**binary**).

Warm Up

For each of the following statements **A** to **D**, answer whether the statement describes analog or digital.

A A thermometer that represents a continuous quantity using the length of a mercury column.

B Accurate and highly reproducible, without any degradation due to duplication or transmission.

C A quantity that changes continuously can be represented using the numbers 0 and 1.

D All information (numbers, text, audio, images, videos, etc.) can be represented in binary.

Explanation

A: Analog

B: Digital

C: Digital

D: Digital

Try

Answer the following questions.

(1) Choose the terms that best fit into the blanks [1] to [5] from the options **A** to **F** below.

> A quantity that is commonly found in our daily lives and changes continuously, such as time or temperature, is called ([1]). In contrast, a quantity that represents a continuously changing value by dividing it into regular intervals and expressing it numerically is called ([2]). Converting analog data to digital data is called ([3]) conversion, while the reverse is called ([4]) conversion.
>
> Furthermore, when handling characters, images, audio, video, and other data on a computer, the data is represented using two numbers, 0 and 1, which is referred to as ([5]).

A Binary **B** Decimal **C** Digital **D** Analog
E A/D **F** D/A

(2) For each of the following statements **A** to **E**, answer whether the statement describes analog or digital.

A Accumulation and duplication of information can be easily achieved.
B Temperature is measured by assessing the height of the mercury column.
C Even if some distortion occurs in the electrical signal due to noise during transmission, it can be easily corrected.
D Quantities that change continuously, such as time or temperature.
E It is possible to comprehensively handle various forms of media.

Exercise

1 Answer the following questions.

(1) What is the term for quantities that can be measured continuously and with fine precision, such as mass or time?
(2) What is the term for the process of converting digital data into analog data?
(3) What is A/D conversion? Explain briefly using the terms “digital” and “analog.”
(4) What is the term for the method of representing numbers using two types of digits (0 and 1)?

2 Choose one statement that is incorrect in regard to digital data from the options **A** to **D** below, and answer using the letter.

A Accumulation and duplication of information can be easily achieved.
B It is possible to efficiently communicate the information.
C It can express quantities that change continuously, such as time and temperature.
D It is possible to combine various types of media.

6-2 Binary and Amount of Information

Point!

1 Bits and Bytes

(1) ([1]**Bit**) : The smallest unit of information, having only two possible states: "0" and "1". One bit is capable of representing two states; for example, "a switch is on or off," "voltage is high or low," "or the orientation of a magnet is north or south."

- Actual information is represented by the arrangement of bits. In general, n bits can represent ([2]2^n) different pieces of information.

<Example> 1 bit : 2 possibilities: 0 or 1
2 bits : 4 possibilities: 00, 01, 10, 11 — $2^2 = 4$ possibilities
3 bits : 8 possibilities: 000, 001, 010, 011, 100, 101, 110, 111 — $2^3 = 8$ possibilities

(2) ([3]**Byte**) : A unit consisting of 8 bits. The notation is ([4]**B**).
1 B = ([5]**8 bits**), which can represent ([6]**256**) possibilities. — $2^8 = 256$ possibilities

(3) Unit of information : The basic unit is 1 B. The unit changes every 2^{10} = ([7]**1,024**) times.
[1] 1 ([8]**KB**) = 1,024 B
[2] 1 ([9]**MB**) = 1,024 KB
[3] 1 ([10]**GB**) = 1,024 MB
[4] 1 ([11]**TB**) = 1,024 GB

2 Decimal and Binary

(1) ([12]**Decimal**) : A method of representing numbers using ten digits from "0" to "9". A number expressed in decimal notation is denoted as ([13]**decimal**).

(2) ([14]**Binary**) : A method of representing numbers using two types of digits, "0" and "1". A number represented in binary is sometimes written with a subscript (2) placed at the bottom right of the number, such as ([15]**binary number**).

(3) Conversion between Decimal and Binary
[1] Conversion from binary to decimal
Starting from the rightmost digit, multiply each digit by 1, 2^1, 2^2, 2^3, ... in order, and then sum the results.
<Example> Convert the binary number $1011_{(2)}$ into decimal form.
$1011_{(2)} = (1 \times 2^0 + 1 \times 2^1 + 0 \times 2^2 + 1 \times 2^3) = 11$

[2] Conversion from decimal to binary
Take a decimal number and keep dividing it by 2 until the quotient becomes 1, then arrange the resulting quotients and remainders in reverse order from the left.
<Example> Represent the number "6" in decimal as a number in binary.

$$\begin{array}{l} 2\overline{)6} \\ 2\overline{)3} \cdots 0 \\ \quad 1 \cdots 1 \end{array}$$

Therefore, 6 = ([17]**$110_{(2)}$**)

Warm Up

Answer the following questions.

(1) [1] Complete the following sentences by filling in the blanks **A** to **E** with the appropriate terms and numbers.

> The smallest unit of information is called a bit, which corresponds to (**A**) digit in binary notation. This makes it possible to represent (**B**) different types of information. Additionally, a group of (**C**) bits is called one (**D**), and is represented as 1 B. For example, 24 bits is (**E**) B.

[2] How many bits of information are required to represent all possible outcomes when throwing a set of dice (one large and one small)?

[3] How many B is 1 MB? Answer in the form of a power of 2.

[4] How many times greater is 4 bits of information compared to 2 bits of information?

(2) [1] Express the following binary numbers in decimal form.

(A) $11010_{(2)}$ (B) $101011_{(2)}$

[2] Express the following decimal number in binary.

(A) 39 (B) 120

Explanation

(1) [1] **A:** A single digit in binary can represent two different values. 1 **B:** 2 **C:** 8
D: Byte **E:** Since 8 bits = 1 B, 24 bits ÷ 8 = 3 B

[2] When you roll two dice (one large and one small), there are 6 × 6 = 36 possible combinations of outcomes.
With 5 bits, you can represent $2^5 = 32$ combinations,
and with 6 bits, you can represent $2^6 = 64$ combinations,
so the needed information is 6 bits

With n bits, it is possible to represent 2^n different pieces of information.

[3] 1 MB = 1,024 KB, and 1 KB = 1,024 B,
therefore, 1 MB = 1,024 KB = 1,024 × 1,024 B = $2^{10} \times 2^{10}$ B = 2^{20} B

[4] The amount of information that can be represented with 2 bits is $2^2 = 4$ possibilities, and the amount of information that can be represented with 4 bits is $2^4 = 16$ possibilities. Therefore, 16 ÷ 4 = 4 times

(2) [1] (A) $11010_{(2)} = 0 \times 1 + 1 \times 2^1 + 0 \times 2^2 + 1 \times 2^3 + 1 \times 2^4$
$= 0 + 2 + 0 + 8 + 16 = 26$

(B) $101011_{(2)} = 1 \times 1 + 1 \times 2^1 + 0 \times 2^2 + 1 \times 2^3 + 0 \times 2^4 + 1 \times 2^5$
$= 1 + 2 + 0 + 8 + 0 + 32 = 43$

[2] (A)

2) 39
2) 19 ⋯1
2) 9 ⋯1
2) 4 ⋯1
2) 2 ⋯0
1 ⋯0

Therefore, $100111_{(2)}$

Add (2) to the bottom right

(B)

2) 120
2) 60 ⋯0
2) 30 ⋯0
2) 15 ⋯0
2) 7 ⋯1
2) 3 ⋯1
1 ⋯1

Therefore, $1111000_{(2)}$

Try

1 Answer the following questions.

(1) Complete the following sentences by filling in the blanks [1] to [3] with the appropriate terms and numbers.

> The numbers we use in our daily lives are expressed in (　[1]　), using digits from 0 to 9. There is also (　[2]　) using the digits 0 and 1, and the information handled by computers primarily utilizes ([2]). However, because this unit is too small and difficult to understand, one byte, which consists of (　[3]　) bits, is often used to represent information.

(2) How many bits are in 5 bytes?

(3) How many different pieces of information can be represented by 1 byte?

(4) How many times is the amount of information in 5 bits compared to the amount of information in 3 bits?

(5) How many MB are in 1 GB? Also, how many B is it? Answer in the form of a power of 2.

(6) How many bits of information are needed to represent the four seasons of spring, summer, autumn, and winter?

(7) How many bits of information are needed to represent a deck of 52 playing cards, excluding the jokers?

(8) How many bits of information are needed to record all possible outcomes when tossing a coin three times?

2 Answer the following questions.

(1) Express the following binary numbers in decimal form.

[1] $110_{(2)}$　　[2] $10100_{(2)}$　　[3] $111001_{(2)}$

(2) Express the following decimal number in binary.

[1] 65　　[2] 106　　[3] 143

♪ **3** How many 4.7 GB DVDs worth of data can be stored on a 1 TB hard disk? Round your answer to the nearest whole number.

Exercise

1 Answer the following questions.

(1) Choose the number or equation that best fit into the blanks [1] to [6] from the options **A** to **J**.

> The amount of information that can be represented by 1 bit is ([1]) possibilities. Additionally, since 1 byte consists of ([2]) bits, the amount of information that can be represented by 1 byte is ([3]) possibilities. For example, 32 bits equal ([4]) bytes. Additionally, 1 KB is expressed as ([5]) B. To calculate how many bits are in 24 KB, you need to calculate the expression ([6]).

A 1	**B** 2	**C** 4	**D** 8
E 256	**F** 1,024	**G** 24 × 1,024	**H** 1,024 ÷ 24
I 24 × 1,024 × 8	**J** 24 × 1,024 ÷ 8		

(2) Choose one option that represents the correct order of the amount of information from the options **A** to **D** below.

A 1 KB < 1 MB < 1 GB < 1 TB
B 1 KB < 1 MB < 1 TB < 1 GB
C 1 TB < 1 KB < 1 MB < 1 GB
D 1 KB < 1 TB < 1 MB < 1 GB

(3) How many different combinations of designs can be created using one 100 yen coin, one 50 yen coin, and one 10 yen coin? Also, how many bits of information does this represent?

(4) How many bits of information are needed to represent 16 directions?

(5) What is the minimum number of bits needed to represent the 47 prefectures?

2 Answer the following questions.

(1) Express the following binary numbers in decimal form.

[1] $101_{(2)}$ [2] $10101_{(2)}$ [3] $101101_{(2)}$

(2) Express the following decimal number in binary.

[1] 13 [2] 128 [3] 138

3 How many photos, each with a size of 2 MB, can be stored on a 32 GB USB flash drive?

6-3 Hexadecimal

Point!

Hexadecimal

(1) ([1]**Hexadecimal**) : A method of representing numbers using the digits 0 to 9 and the alphabet letters A to F. A number represented in hexadecimal is sometimes written with a subscript (16) placed at the bottom right of the number. This is known as a ([2]**hexadecimal number**).

(2) Correspondence Between Decimal, Binary, and Hexadecimal

Decimal	Binary	Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8

Decimal	Binary	Hexadecimal
9	1001	9
10	1010	([3]**A**)
11	1011	([4]**B**)
12	1100	([5]**C**)
13	1101	([6]**D**)
14	1110	([7]**E**)
15	1111	([8]**F**)
16	10000	([9]**10**)

(3) Conversion between Decimal, Binary, and Hexadecimal

[1] Conversion from binary to hexadecimal

Separate the binary number into groups of 4 digits starting from the least significant bit, convert each group into a hexadecimal value, and arrange these values in sequence.

<Example> Convert the binary number $10011010_{(2)}$ to a hexadecimal number.

Separate into groups of 4 digits starting from the least significant bit: $1001/1010_{(2)}$

$1001_{(2)}$ is $(9_{(16)})$, $1010_{(2)}$ is $(A_{(16)})$, so $10011010_{(2)} = (9A_{(16)})$

[2] Conversion from hexadecimal to binary

Convert each digit of the hexadecimal number into a 4-bit binary number, and arrange those numbers sequentially.

<Example> Express hexadecimal $A4_{(16)}$ in binary.

$A_{(16)}$ is $(1010_{(2)})$, $4_{(16)}$ is $(0100_{(2)})$, so $A4_{(16)} = (10100100_{(2)})$

[3] Conversion from hexadecimal to decimal

Convert the hexadecimal number to a binary number, and then convert that binary number to a decimal number.

<Example> Express the hexadecimal $C6_{(16)}$ in decimal format.

First, to convert the hexadecimal $C6_{(16)}$ to binary, $C_{(16)}$ is $(1100_{(2)})$,

$6_{(16)}$ is $(0110_{(2)})$, so $C6_{(16)} = (11000110_{(2)})$

Next, to convert the binary number $11000110_{(2)}$ into a decimal number,

$11000110_{(2)} = (0 \times 1 + 1 \times 2^1 + 1 \times 2^2 + 0 \times 2^3 + 0 \times 2^4 + 0 \times 2^5 + 1 \times 2^6 + 1 \times 2^7)$

$= 198$

Warm Up

Answer the following questions.

(1) Convert the following binary numbers to hexadecimal, and the hexadecimal numbers to binary.

[1] $11011011_{(2)}$ [2] $11110110_{(2)}$ [3] $9E_{(16)}$

(2) Convert the following hexadecimal numbers to decimal.

[1] $A5_{(16)}$

Explanation

(1) [1] Separate the binary number into groups of four digits starting from the least significant bit, then convert each group into a hexadecimal number and list them sequentially.
$1101_{(2)}$ is $D_{(16)}$, $1011_{(2)}$ is $B_{(16)}$, so $11011011_{(2)} = \underline{DB_{(16)}}$

[2] $1111^{(2)}$ is $F_{(16)}$, $0110_{(2)}$ is $6_{(16)}$, so $11110110_{(2)} = \underline{F6_{(16)}}$

[3] Convert each digit into a 4-bit binary number and list them sequentially.
$9_{(16)}$ is $1001_{(2)}$, $E_{(16)}$ is $1110_{(2)}$, so $9E_{(16)} = \underline{10011110_{(2)}}$

(2) Convert from hexadecimal to binary, and then to decimal.

[1] $A_{(16)}$ is $1010_{(2)}$, $5_{(16)}$ is $0101_{(2)}$, so $A5_{(16)} = 10100101_{(2)}$
Therefore, $10100101_{(2)} = 1 \times 1 + 0 \times 2^1 + 1 \times 2^2 + 0 \times 2^3 + 0 \times 2^4 + 1 \times 2^5 + 0 \times 2^6 + 1 \times 2^7 = \underline{165}$

Try

Answer the following questions.

(1) Convert the following binary numbers to hexadecimal, and the hexadecimal numbers to binary.

[1] $11010101_{(2)}$ [2] $01110110_{(2)}$ [3] $C5_{(16)}$ [4] $BB_{(16)}$

(2) Convert the following hexadecimal numbers to decimal.

[1] $31_{(16)}$ [2] $C7_{(16)}$ [3] $9F_{(16)}$ [4] $AB_{(16)}$

Exercise

1 Answer the following questions.

(1) Convert the following binary numbers to hexadecimal, and the hexadecimal numbers to binary.

[1] $01010011_{(2)}$ [2] $01101100_{(2)}$ [3] $10100111_{(2)}$ [4] $11001001_{(2)}$
[5] $2B_{(16)}$ [6] $F1_{(16)}$ [7] $EE_{(16)}$ [8] $DB_{(16)}$

(2) Convert the following hexadecimal numbers to decimal.

[1] $17_{(16)}$ [2] $C2_{(16)}$ [3] $3D_{(16)}$ [4] $AF_{(16)}$

6-4 Digital Representation of Characters

Point!

Digital Representation of Characters

(1) ([1]**Character code**) : A unique numerical value assigned to each character, symbol, etc.

(2) ([2]**Character code system**) : Summarizes the correspondence between characters and their respective character codes.

[1] ([3]**ASCII code**) : Only alphanumeric characters, symbols, and control characters (symbols used to control the computer). No Japanese characters (*kanji*, *hiragana*, or *katakana*).

		Most Significant 4 Bits								
		Binary	0000	0001	0010	0011	0100	0101	0110	0111
Least Significant 4 Bits	Binary	Hexadecimal	0	1	2	3	4	5	6	7
	0000	0	Symbols for Controlling Computers (Omission)		SP	0	@	P	`	p
	0001	1			!	1	A	Q	a	q
	0010	2			”	2	B	R	b	r
	0011	3			#	3	C	S	c	s
	0100	4			$	4	D	T	d	t
	0101	5			%	5	E	U	e	u
	0110	6			&	6	F	V	f	v
	0111	7			'	7	G	W	g	w
	1000	8			(	8	H	X	h	x
	1001	9			)	9	I	Y	i	y
	1010	A			*	:	J	Z	j	z
	1011	B			+	;	K	[	k	{
	1100	C			,	<	L	¥	l	|
	1101	D			-	=	M	]	m	}
	1110	E			.	>	N	^	n	~
	1111	F			/	?	O	_	o	DEL

<Example> Convert the string “Hello” into a binary character code and a hexadecimal character code. Combine the most significant 4 bits and the least significant 4 bits.

String	H	e	l	l	o
Character code (binary)	01001000	01100101	01101100	01101100	01101111
Character Code (hexadecimal)	48	65	6C	6C	6F

[2] ([7]**Unicode**) : A character code standard that consolidates characters from around the world into a single character code. Due to differences in code assignments, there are variations such as UTF-8 and UTF-16.

(3) ([8]**Encoding**) : Representing a string with character codes. The opposite is known as ([9]**decoding**).

(4) ([10]**Character corruption**) : A phenomenon that occurs due to mismatched encoding and decoding methods.

(5) ([11]**Font**) : The shape data of characters corresponding to character codes.
<Examples> Sans-serif, Serif, Semi-cursive, etc.

- To display characters on a computer screen or printer output, two elements are required: ([12]**character code**) and ([13]**font**).

Warm Up

Answer the following questions.

(1) Choose the numbers or terms that best fit into the blanks [1] to [5] from the options **A** to **H** below.

> Characters and symbols can also be represented by combinations of 0s and 1s. A system that maps characters and symbols to binary or hexadecimal is called a ([1]). There are several types within the ([1]), including ([2]), which accommodates the writing systems of countries around the world.
>
> Alphanumeric characters and symbols are represented by ([3]) byte, 1 byte can represent ([4]) types of characters.

A 1 **B** 2 **C** 128 **D** 256

E Character code **F** Character corruption **G** Unicode **H** ASCII code

(2) Use the character code table on the right to answer the following questions.

[1] Use binary to represent the character code for "M".

[2] Use hexadecimal to represent the character code for "MILK".

[3] What is the character string corresponding to the character code "$4C6F7665_{(16)}$"?

		Most Significant 4 Bits								
		Binary	0000	0001	0010	0011	0100	0101	0110	0111
Least Significant 4 Bits	Binary	Hexadecimal	0	1	2	3	4	5	6	7
	0000	0	Symbols for Controlling Computers (Omission)		SP	0	@	P	`	p
	0001	1			!	1	A	Q	a	q
	0010	2			"	2	B	R	b	r
	0011	3			#	3	C	S	c	s
	0100	4			$	4	D	T	d	t
	0101	5			%	5	E	U	e	u
	0110	6			&	6	F	V	f	v
	0111	7			'	7	G	W	g	w
	1000	8			(	8	H	X	h	x
	1001	9			)	9	I	Y	i	y
	1010	A			*	:	J	Z	j	z
	1011	B			+	;	K	[	k	{
	1100	C			,	<	L	¥	l	|
	1101	D			-	=	M	]	m	}
	1110	E			.	>	N	^	n	~
	1111	F			/	?	O	_	o	DEL

Explanation

(1) [1] **E** [2] **G** [3] **A** [4] **B** [5] **D**

(2) [1] From the character code table, the most significant 4 bits of "M" are 0100 and the least significant 4 bits are 1101; therefore, $01001101_{(2)}$

[2] According to the character code table, the character codes corresponding to "M, I, L, K" are 4D, 49, 4C, and 4B, respectively, in the order of most significant to least significant. Connecting these character codes results in $4D494C4B_{(16)}$

[3] Consider the character code as separated into "4C/6F/76/65".
"4C" corresponds to the character "L" because it is located at the intersection of the most significant 4 and the least significant C.
Similarly, "6F" corresponds to "o", "76" to "v", and "65" is "e", so the resulting character string is Love

Try

1 Answer the following questions.

(1) Complete the following sentences by filling in the blanks [1] to [4] with the appropriate terms and numbers.

> The rule for how characters and symbols are represented in binary or hexadecimal is called a (　[1]　). Alphanumeric characters and symbols are represented by (　[2]　) byte. In principle, 1 byte can represent (　[4]　) types of characters, and 2 bytes can represent 65,536 types of characters.

(2) Briefly explain the characteristics of the character code system Unicode.

(3) Use the character code table below to answer the following questions.

		Most Significant 4 Bits								
		Binary	0000	0001	0010	0011	0100	0101	0110	0111
Least Significant 4 Bits	Binary	Hexadecimal	0	1	2	3	4	5	6	7
	0000	0	Symbols for Controlling Computers (Omission)		SP	0	@	P	`	p
	0001	1			!	1	A	Q	a	q
	0010	2			”	2	B	R	b	r
	0011	3			#	3	C	S	c	s
	0100	4			$	4	D	T	d	t
	0101	5			%	5	E	U	e	u
	0110	6			&	6	F	V	f	v
	0111	7			'	7	G	W	g	w
	1000	8			(	8	H	X	h	x
	1001	9			)	9	I	Y	i	y
	1010	A			*	:	J	Z	j	z
	1011	B			+	;	K	[	k	{
	1100	C			,	<	L	¥	l	|
	1101	D			-	=	M	]	m	}
	1110	E			.	>	N	^	n	~
	1111	F			/	?	O	_	o	DEL

[1] What is the character code for “a” in binary?

[2] What is the symbol corresponding to the character code “$00111110_{(2)}$”?

[3] What is the symbol corresponding to the character code “$5C_{(16)}$”?

[4] What is the character string corresponding to the character code “$486172696E657A756D69_{(16)}$”?

(4) Calculate the following data size.

[1] The data size when “Welcome” is entered.

(5) Choose the shape data of characters corresponding to character codes from the options **A** to **C** below.

A Font　　**B** Environment-dependent characters　　**C** *Emoji*

Exercise

1 Answer the following questions.

(1) Answer the following questions.

[1] What is the term for a unique numerical value assigned to each character, symbol, etc.?

[2] How many types of characters can be represented with 1 byte?

[3] What is the term for representing a string with character codes?

[4] What is the term for a design that corresponds to a character code and shares the same characteristics?

(2) Briefly explain the characteristics of the character code system ASCII.

(3) Use the character code table below to answer the following questions.

		Most Significant 4 Bits								
		Binary	0000	0001	0010	0011	0100	0101	0110	0111
Least Significant 4 Bits	Binary	Hexadecimal	0	1	2	3	4	5	6	7
	0000	0	Symbols for Controlling Computers (Omission)		SP	0	@	P	`	p
	0001	1			!	1	A	Q	a	q
	0010	2			”	2	B	R	b	r
	0011	3			#	3	C	S	c	s
	0100	4			$	4	D	T	d	t
	0101	5			%	5	E	U	e	u
	0110	6			&	6	F	V	f	v
	0111	7			'	7	G	W	g	w
	1000	8			(	8	H	X	h	x
	1001	9			)	9	I	Y	i	y
	1010	A			*	:	J	Z	j	z
	1011	B			+	;	K	[	k	{
	1100	C			,	<	L	¥	l	|
	1101	D			-	=	M	]	m	}
	1110	E			.	>	N	^	n	~
	1111	F			/	?	O	_	o	DEL

[1] What is the character code for “Y” in binary?

[2] What is the character corresponding to the character code “$01101111_{(2)}$”?

[3] What is the character corresponding to the character code “$72_{(16)}$”?

[4] What is the character string corresponding to the character code “$5468616E6B596F7521_{(16)}$”?

(4) Calculate the following data size.

[1] The data size when “HappyBirthday” is entered

6-5 Numerical Calculations [1]

Point!

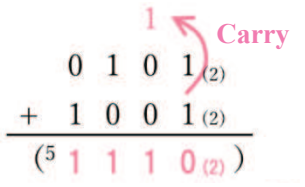
Addition and Subtraction of Binary Numbers

Addition and subtraction in binary numbers are performed digit by digit, just like in decimal numbers.

(1) Binary addition : In binary addition, the place value carries over by one position when the sum reaches 2.

$0_{(2)} + 0_{(2)} = ({}^{1}0_{(2)})$, $0_{(2)} + 1_{(2)} = ({}^{2}1_{(2)})$, $1_{(2)} + 1_{(2)} = ({}^{3}10_{(2)})$, $1_{(2)} + 1_{(2)} + 1_{(2)} = ({}^{4}11_{(2)})$

<Example> Addition of binary numbers $0101_{(2)} + 1001_{(2)}$

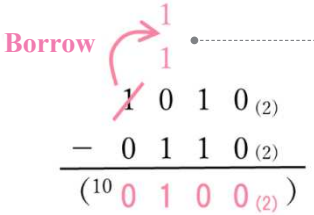


When two 1s are added together, a carry of 1 is generated to the next higher digit.

(2) Binary subtraction : In binary subtraction, when the minuend is insufficient, a value of 2 is borrowed from the next higher digit.

$0_{(2)} - 0_{(2)} = ({}^{6}0_{(2)})$, $1_{(2)} - 0_{(2)} = ({}^{7}1_{(2)})$, $1_{(2)} - 1_{(2)} = ({}^{8}0_{(2)})$, $10_{(2)} - 1_{(2)} = ({}^{9}1_{(2)})$

<Example> Subtraction of binary numbers $1010_{(2)}$ - $0110_{(2)}$



When 1 is borrowed from the higher digit, it becomes two 1s in the lower digit.

Warm Up

Perform the following binary calculations.

(1) $0101_{(2)} + 1010_{(2)}$

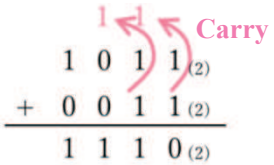
(2) $1011_{(2)} + 0011_{(2)}$

(3) $1011_{(2)} - 0101_{(2)}$

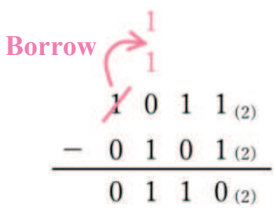
(4) $1101_{(2)} - 1010_{(2)}$

Explanation

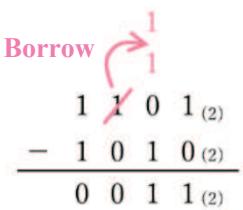
(1)
$$\begin{array}{r} 0\ 1\ 0\ 1_{(2)} \\ +\ 1\ 0\ 1\ 0_{(2)} \\ \hline 1\ 1\ 1\ 1_{(2)} \end{array}$$
Therefore, $\underline{1111_{(2)}}$

(2)


Therefore, $\underline{1110_{(2)}}$

(3)


Therefore, $\underline{0110_{(2)}}$

(4)


Therefore, $\underline{0011_{(2)}}$

Try

Perform the following binary calculations.

(1) $0101_{(2)} + 0110_{(2)}$

(2) $1101_{(2)} + 0010_{(2)}$

(3) $1101_{(2)} - 0010_{(2)}$

(4) $1010_{(2)} - 0101_{(2)}$

Exercise

1 Perform the following binary calculations.

(1) $0011_{(2)} + 1100_{(2)}$

(2) $1001_{(2)} + 0101_{(2)}$

(3) $0111_{(2)} + 0001_{(2)}$

(4) $1010_{(2)} + 0101_{(2)}$

(5) $1101_{(2)} - 1100_{(2)}$

(6) $1011_{(2)} - 0110_{(2)}$

(7) $1001_{(2)} - 0110_{(2)}$

(8) $1010_{(2)} - 1010_{(2)}$

6-6 Numerical Calculations [2]

Point!

1 Representation of Negative Numbers Using Complements

([1]**Complement**) : The smallest number which, when added to a given natural number, produces a carry to the next higher digit. Complements are used for representing negative numbers in computers.

[1] ([2]**10's complement**) : In decimal, the smallest number which, when added, produces a carry to the next higher digit.

<Examples> The 10's complement of 71 is ([3]**29**), and the 10's complement of 635 is ([4]**365**).

[2] ([5]**2's complement**) : In binary, the smallest number which, when added, produces a carry to the next higher digit.

<How to calculate the 2's complement>

For each digit of the original binary number, invert 0s to 1s and 1s to 0s, then add 1 at the end to mechanically obtain the result.

<Example> The 2's complement of $0101_{(2)}$

By inverting each digit from 0 to 1 and from 1 to 0, it becomes ([6]**$1010_{(2)}$**). Finally, by adding 1, it becomes ([7]**$1011_{(2)}$**).

2 Subtraction Using Complements

A computer uses subtraction by performing addition with complements.

<Subtraction process>

1) Find the ([8]**complement**) of the subtrahend.

2) Use the complement to perform ([9]**addition**).

3) Ignore the leading digits of the calculation result and provide the answer.

<Example 1>

Subtraction of "8 − 6" in decimal

1) The 10's complement of the decimal number 6 is ([10]**4**)

2) 8 − 6 ⇒ 8 ([11]+ 4)
= ([12]**12**)

3) Ignoring the leading digit, ([13]2)

<Example 2>

Subtraction of "$1000_{(2)}$ − $0111_{(2)}$" in binary

1) The 2's complement of the binary number $0111_{(2)}$ is ([14]$1001_{(2)}$)

2) $1000_{(2)}$ − $0111_{(2)}$ ⇒ $1000_{(2)}$ ([15]+ $1001_{(2)}$)
= ([16]$10001_{(2)}$)

3) Ignoring the leading digit, ([17]$0001_{(2)}$)

Warm Up

Answer the following questions.

(1) What is the complement of the following binary number?

[1] $1001_{(2)}$ [2] $0100_{(2)}$

(2) Use complements to perform the following binary subtractions.

[1] $1100_{(2)} - 0111_{(2)}$ [2] $1110_{(2)} - 1001_{(2)}$

Explanation

(1) [1] By inverting each digit from 0 to 1 and from 1 to 0, it becomes $0110_{(2)}$.
Finally, by adding 1, it becomes $\underline{0111}_{(2)}$.

[2] By inverting each digit from 0 to 1 and from 1 to 0, it becomes $1011_{(2)}$.
Finally, by adding 1, it becomes $\underline{1100}_{(2)}$. Adding 1 causes a place value to carry over.

(2) [1] The complement of the binary number $0111_{(2)}$ is $1001_{(2)}$, so
$1100_{(2)} - 0111_{(2)} \Rightarrow 1100_{(2)} + 1001_{(2)} = 10101_{(2)}$
By ignoring the leading digit, $\underline{0101}_{(2)}$

[2] The complement of the binary number $1001_{(2)}$ is $0111_{(2)}$, therefore,
$1110_{(2)} - 1001_{(2)} \Rightarrow 1110_{(2)} + 0111_{(2)} = 10101_{(2)}$
By ignoring the leading digit, $\underline{0101}_{(2)}$

Try

Answer the following questions.

(1) What is the complement of the following binary number?

[1] $0101_{(2)}$ [2] $1101_{(2)}$ [3] $10110001_{(2)}$ [4] $01001100_{(2)}$

(2) Use complements to perform the following binary subtractions.

[1] $1101_{(2)} - 0110_{(2)}$ [2] $1010_{(2)} - 0111_{(2)}$ [3] $1101_{(2)} - 1010_{(2)}$

Exercise

1 Answer the following questions.

(1) What is the complement of the following binary number?

[1] $0111_{(2)}$ [2] $1011_{(2)}$ [3] $10011011_{(2)}$ [4] $11000110_{(2)}$

(2) Use complements to perform the following binary subtractions.

[1] $1010_{(2)} - 0110_{(2)}$ [2] $1011_{(2)} - 1001_{(2)}$ [3] $1100_{(2)} - 0010_{(2)}$

6-7 Digitalization of Sound

Point!

1 Sound

(1) Sound : A phenomenon transmitted through the vibration of air. Waveform ([1]**analog**) data.
- For audio data on CDs, etc., sound is converted into ([2]**digital**) data.

(2) ([3]**Frequency**) : The number of waves contained in one second.
The unit is ([4]**hertz**) [Symbol: ([5]**Hz**)].

(3) ([6]**Period**) : The time it takes for one wave to propagate. The unit is ([7]**seconds**).

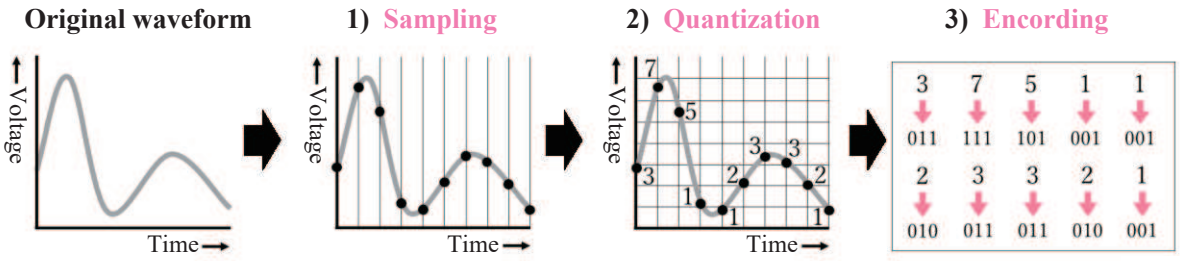


2 Digitalization of Sound

(1) ([8]**Pulse Code Modulation (PCM) method**) : A method for digitizing audio analog data.
Convert sound information into binary code.

<Steps for digitizing sound using the PCM method>

1) ([9]**Sampling**) : The horizontal axis (time) is divided at regular intervals, and the wave height (voltage strength) is extracted. The extracted points are referred to as ([10]**sample points**).
   - ([11]**Sampling period**) : The time interval used to divide when sampling.
   - ([12]**Sampling frequency**) : The number of samples taken per second.
2) ([13]**Quantization**) : The vertical axis (voltage) is divided at regular intervals, and the wave heights obtained from sampling are converted to the nearest values on the vertical scale.
   - ([14]**Quantization bit depth**) : This determines the number of levels into which the range is divided during quantization.
3) ([15]**Encoding**) : Express the quantized values in binary form.



(2) Digitization and Data Volume
As the sampling frequency and the quantization bit depth ([16]**increase**), the sound becomes closer to the original analog waveform. This causes the sound quality to ([17]**improve**); however, the amount of data also ([18]**increases**).

(3) ([19]**Sampling theorem**) : If the sampling frequency exceeds ([20]**twice**) the highest frequency contained in the original analog waveform, the waveform of the original analog signal can be accurately reconstructed from the digitized data.

3 Amount of Sound Data

(1) Channels : The number of signals used when transmitting sound. A method of playback using a single signal is called ([21]**monaural**), and a method of playback using two different signals is called ([22]**stereo**).

(2) How to Calculate the Amount of Audio Data
Data amount [bits] per second = ([23]**Sampling frequency × Quantization bit depth × Number of channels**)

Warm Up

Answer the following questions.

(1) Choose the one appropriate procedure for digitizing an analog sound signal from the options **A** to **D**.

A Sampling → Quantization → Encoding
B Encoding → Quantization → Sampling
C Quantization → Sampling → Encoding
D Quantization → Encoding → Sampling

(2) Calculate the data amount for 1 second in kilobytes (KB) when digitizing music at a sampling frequency of 44,100 Hz, a quantization bit depth of 16 bits, and in 2-channel stereo. However, assume that 1 KB = 1,000 B, and round off to the nearest whole number.

(3) Calculate the data amount for 1 minute in megabytes (MB) when digitizing music at a sampling frequency of 44.1 kHz, a quantization bit depth of 24 bits, and in 2-channel stereo. However, assume that 1 KB = 1,000 B, and 1 MB = 1,000 KB, and round off to the nearest whole number.

Explanation

(1) **A**

(2) Data amount [bits] per second = Sampling frequency × Quantization bit depth × Number of channels
Therefore, the amount of data per second is 44,100 [Hz] × 16 [bits] × 2 [channels] = 1,411,200 [bits].
Since 1 B = 8 bits, 1,411,200 [bits] ÷ 8 = 176,400 [B]
1 KB = 1,000 B, so 176,400 [B] ÷ 1,000 = 176.4 ≈ 176 [KB] 176 KB

(3) 44.1 kHz = 44.1 × 1,000 = 44,100 Hz, and since 2 minutes = 120 seconds, the data amount per minute is 44,100 [Hz] × 24 [bits] × 2 [channels] × 60 [seconds] = 127,008,000 [bits]
Since 1 B = 8 bits, 127,008,000 [bits] ÷ 8 = 15,876,000 [B].
1 KB = 1,000 B, 1 GB = 1,000 MB, therefore,
15,876,000 [B] ÷ 1,000 ÷ 1,000 = 15.876 [MB] ≈ 16 [MB] 16 MB

Try

1 Answer the following questions.

(1) Choose the term that best fits into the blanks [1] to [10] in the following sentences from the options **A** to **M** below, and answer using the letters.

> Sound is a phenomenon that is transmitted through the vibrations of air. The number of waves contained in one second is called ([1]) and is expressed in units of ([2]). The time it takes for one wave to propagate is called the ([3]), and it is expressed in seconds. These vibrations are ([4]) data. In contrast, sound information stored on DVDs, CDs, etc. is ([5]) data. Next, follow these steps to convert the data from ([4]) to ([5]).
>
> First, the process of dividing sound waves at regular time intervals and extracting the amplitude values (voltage) for each interval is called ([6]). Next, divide the vertical axis (voltage) at regular intervals and determine the step values. This is called ([7]), and the component that determines the quantization levels is called ([8]). Typically, with n bits, there are 2^n levels. Finally, ([9]) refers to the representation of values obtained through ([7]) in binary form. The method of converting information such as sound into binary codes by replacing numerical values in binary with two different voltage levels (high and low) is called the ([10]) method.

A	Encoding	**B**	Sampling	**C**	Quantization	**D**	Frequency	**E**	Period
F	Analog	**G**	Digital	**H**	Sampling period	**I**	Quantization bit depth		
J	PCM	**K**	MIDI	**L**	Hertz	**M**	Second		

(2) Choose the one that enables waveform restoration closest to the original analog signal when digitizing an analog audio signal from the options **A** to **D**.

A Increase the sampling frequency and increase the quantization bit depth.
B Increase the sampling frequency and decrease the quantization bit depth.
C Decrease the sampling frequency and increase the quantization bit depth.
D Decrease the sampling frequency and decrease the quantization bit depth.

2 Answer the following questions. In these questions, assume that 1 KB = 1,000 B, and 1 MB = 1,000 KB.

(1) What is the data amount for 1 second in kilobytes (KB) when digitizing music at a sampling frequency of 44,100 Hz, a quantization bit depth of 16 bits, and in 1-channel monaural? Round off to the nearest whole number in your answer.

(2) What is the data amount for 1 minute in KB when digitizing music at a sampling frequency of 192 kHz, a quantization bit depth of 24 bits, and in 2-channel stereo?

(3) What is the data amount for 3 minutes in MB when digitizing music at a sampling frequency of 44.1 kHz, a quantization bit depth of 16 bits, and in 2-channel stereo? If the length of one song is 3 minutes, approximately how many songs can be stored on a 650 MB CD? Round off to the nearest whole number in your answer.

Exercise

1 Answer the following questions.

(1) Choose the term that best fits into the blanks [1] to [6] in the following sentences from the options **A** to **I** below, and answer using the letters.

> Sound is analog data in waveforms that are transmitted through the vibration of air. The process of capturing sound waveforms on a computer begins with segmenting the waveforms at regular time intervals. This process is called ([1]). The number of partitions per second is called the ([1]) frequency. Next, convert and approximate the segmented wave heights into a continuous bar graph. This process is called ([2]). At this time, the value indicating how many levels the numbers are expressed in is called ([3]). Finally, perform ([4]), which uses 0s and 1s to express the value in binary after ([2]). This method of converting sound into digital data is called ([5]).
>
> In the case of digitizing sound, digital data converted from analog data will approximate the original waveform more closely if the ([1]) frequency is higher. Furthermore, the greater the ([3]), the more accurately the waveform can be represented numerically. However, as a result, the amount of data becomes ([6]).

A Encoding	**B** Sampling	**C** Quantization	**D** Larger	**E** Smaller
F Sampling period	**G** Quantization bit depth	**H** PCM method	**I** MIDI format	

(2) The following statements **A** to **D** are about the sampling frequency and the number of quantization levels. Mark "○" if the statement is correct, and "×" if it is incorrect.

A Sampling frequency refers to the number of times sampling occurs in one second.

B The greater the number of levels during quantization, the larger the amount of data becomes.

C If you want to approximate the original analog waveform more closely, it is effective to increase the sampling frequency and reduce the number of levels during quantization.

D If the maximum frequency of the original analog waveform is 40 Hz, you should digitize it with a sampling frequency greater than 80 Hz.

2 Answer the following questions. In these questions, assume that 1 KB = 1,000 B, and 1 MB = 1,000 KB.

(1) What is the data amount for 10 seconds in kilobytes (KB) when digitizing music at a sampling frequency of 44,100 Hz, a quantization bit depth of 16 bits, and in 1-channel monaural?

(2) What is the data amount for 1 minute in MB when digitizing music at a sampling frequency of 96 kHz, a quantization bit depth of 24 bits, and in 2-channel stereo? Round off to the nearest whole number in your answer.

6-8 Digitization of Images

Point!

1 Digitization of Images

(1) ([1]**Pixel**) : The smallest unit that composes an image.
A digital image is represented by an arrangement of pixels.

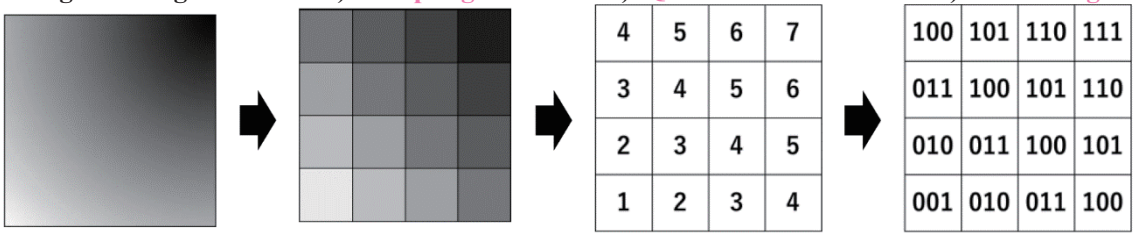
(2) Procedure for Digitization of Images
Images captured by digital cameras or image scanners are digitized and imported into a computer through the following process.

1) ([2]**Sampling**) : The image is divided into pixels, and the representative brightness is extracted.
- ([3]**Resolution**) : The degree of fineness of pixels when sampling.
The unit of resolution is ([4]**dpi**).
It is sometimes expressed as vertical pixels × horizontal pixels.

2) ([5]**Quantization**) : Converts the brightness of each pixel into numerical values divided into several levels.
- ([6]**Gradation**) : The level value that represents the intensity of color per pixel. The gradation is determined by the number of bits allocated to each pixel.
<Example> At 256 gradients (8 bits), it represents 256 levels from 0 to 255.

3) ([7] **Encoding**) : Expressing quantized values in binary numbers (0 and 1).

Original Image → **1) Sampling** → **2) Quantization** → **3) Encording**



2) Quantization			
4	5	6	7
3	4	5	6
2	3	4	5
1	2	3	4

3) Encording			
100	101	110	111
011	100	101	110
010	011	100	101
001	010	011	100

(3) Digitization and Data Volume
[1] Data amount of an image [bits]
= ([8] **Number of pixels [vertical pixels × horizontal pixels) × Number of bits for color information**)
[2] The ([9]**higher**) the values of resolution and gradation, the smoother the image obtained and the image quality ([10]**improves**), but the amount of data ([11]**increases**).

(4) Raster Format and Vector Format
[1] ([12]**Raster format**) : An arrangement of pixels used to represent text or shapes. It appears jagged when enlarged (this appearance is called ([13]**jaggies**)).
Images are drawn using ([14]**painting software**).
[2] ([15]**Vector format**) : A representation that includes information about coordinates of points composing the image, as well as the angles and thickness of the lines connecting them. Images are drawn using ([16]**drawing software**).

2 Color Representation

(1) ([17]**Three primary colors of light**) : The three colors of red, green, and blue. When these colors of light are mixed, the brightness increases and approaches ([18]**white**).

(2) ([19]**Three primary colors of pigment**) : The three colors of cyan, magenta, and yellow. Mixing these colors will darken them and bring them closer to ([20]**black**).

(3) ([21]**Additive color mixing**) : A method of representing colors by combining the three primary colors of light. Used in televisions and computer displays.

(4) ([22]**Subtractive color mixing**) : A method of representing colors by combining the three primary colors of pigment. Used in color printers.

(5) ([23]**24-bit full color**) : Each color of red, green, and blue is represented in 256 levels (2^8 = 8 bits) × 3 = 24 bits.

Warm Up

Answer the following questions.

(1) Rearrange the following steps **A** to **C** in the order of converting an image into digital data.
- **A** Replace the brightness levels of each region with integer values according to a fixed rule.
- **B** Divide the image into evenly spaced grid sections.
- **C** Represent integers in binary.

(2) Choose all statements that correctly describe the digital representation of images from the options **A** to **D**, and answer using the letters.
- **A** The raster format appears as having noticeably jagged edges when the image is enlarged.
- **B** The raster format is suitable for expressing images with clear contours, such as logos.
- **C** Vector format represents images by incorporating information such as the coordinates of points and the thickness of lines.
- **D** Painting software is used to create vector graphics.

(3) What is the data amount of a 24-bit full-color image with a resolution of 1,280 × 720 pixels in megabytes? For this question, assume that 1 KB = 1,000 B and 1 MB = 1,000 KB, and round your answer to two decimal places.

Explanation

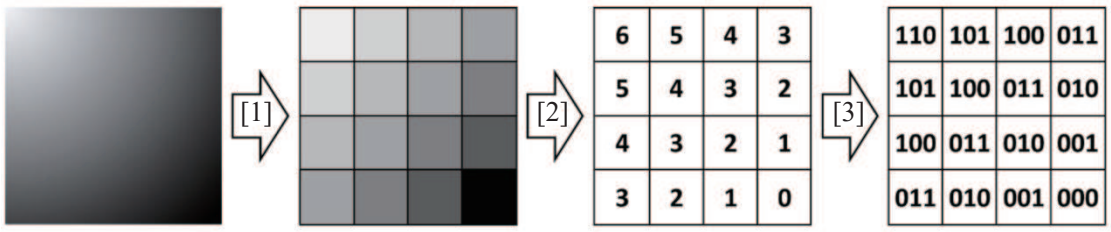
(1) **B → A → C**

(2) **A**, **C**

(3) The amount of data in the image is,
1,280 × 720 × 24 [bits] = 22,118,400 [bits]
Since 1 B equals 8 bits, 22,118,400 ÷ 8 = 2,764,800 [B].
1 KB = 1,000 B, 1 MB = 1,000 KB, therefore,
2,764,800 [B] ÷ 1,000 ÷ 1,000 = 2.7648 [MB] ≈ 2.76 [MB] 2.76 MB

Try

1 The following diagram illustrates the process of converting a monochrome image into digital data. Choose the appropriate operations for steps **[1]** to **[3]** from [Group A], and then choose the name of each operation from [Group B]. Answer using the numbers.



[Group A] **A** Replace the brightness levels of each region with integer values according to a fixed rule.
B Divide the image into evenly spaced grid sections.
C Represent integers in binary.

[Group B] **A** Quantization **B** Encoding **C** Sampling

2 Answer the following questions.

(1) Complete the following sentences by filling in the blanks [1] and [2] with the appropriate term and number.

> To represent colors, a computer display uses a combination of dots in red, green, and blue. These are known as the ([1]). The intensity of each of the primary colors is expressed in gradations. When representing one color with 8 bits, there are ([2]) gradients.

(2) Complete the following sentences by filling in the blanks [1] and [2] with the appropriate terms.

> Images processed by computers can be divided into two formats: the ([1]) format, which treats the image as a collection of tiny dots, and the ([2]) format, which uses a combination of mathematical expressions to represent the coordinates of points, lines and surfaces connecting them, and color information.

(3) Choose one statement that is incorrect as an explanation of the digital representation of images from the options **A** to **D**.

A The unit dpi represents the degree of pixel density.
B The number of gradations depends on the number of bits allocated per pixel.
C The smaller the resolution and gradation values, the smoother the image obtained.
D Gradation indicates the number of steps or levels of color or brightness.

3 Answer the following questions. In these questions, assume that 1 KB = 1,000 B, and 1 MB = 1,000 KB.

(1) What is the data amount of an image in kilobytes with a resolution of 800 × 600 pixels, where each pixel has 8 bits?

(2) What is the data amount of a 24-bit full-color image with a resolution of 720 × 480 pixels in megabytes? Round your answer to two decimal places.

(3) What is the data amount of a 24-bit full-color image with a resolution of 3,820 × 2,160 pixels in megabytes? Round your answer to two decimal places.

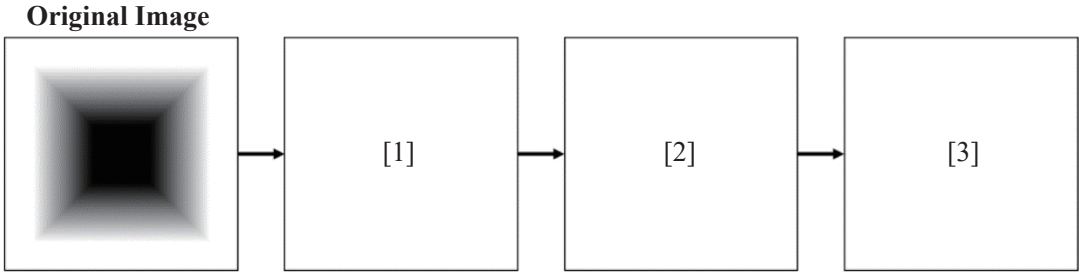
Exercise

1 Answer the following questions.

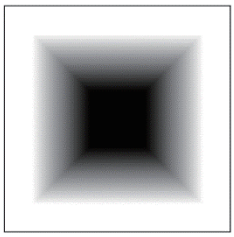
(1) Choose the correct flow for digitizing an image captured by a digital camera from the options **A** to **D**, and answer using the letter.

A Quantization → Encoding → Sampling
B Quantization → Sampling → Encoding
C Encoding → Quantization → Sampling
D Sampling → Quantization → Encoding

(2) The following diagram illustrates the process of converting the original image into digital data. Choose the diagram that best fits into the blanks [1] to [3] from the options **A** to **D**, and answer using the letters.



A



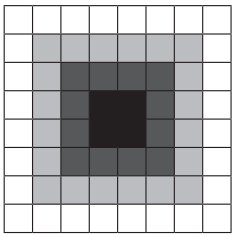
B

0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	0
0	1	2	2	2	2	1	0
0	1	2	3	3	2	1	0
0	1	2	3	3	2	1	0
0	1	2	2	2	2	1	0
0	1	1	1	1	1	1	0
0	0	0	0	0	0	0	0

C

00	00	00	00	00	00	00	00
00	01	01	01	01	01	01	00
00	01	10	10	10	10	01	00
00	01	10	11	11	10	01	00
00	01	10	11	11	10	01	00
00	01	10	10	10	10	01	00
00	01	01	01	01	01	01	00
00	00	00	00	00	00	00	00

D



2 Choose the term that best fits into the blanks [1] to [3] in the following sentences from the options **A** to **D** below.

> In color printers, a variety of colors are expressed by combining the three colors of cyan, magenta, and yellow, which are the ([1]). When these colors are mixed together, they approach the color ([2]). On the other hand, computer displays represent all colors by combining red, green, and blue, which are the ([3]).

A Three primary colors of light
B Three primary colors of pigment
C White
D Black

3 Answer the following questions. In these questions, assume that 1 KB = 1,000 B, and 1 MB = 1,000 KB.

(1) What is the data amount of an image in kilobytes with a resolution of 1,080 × 720 pixels, where each pixel has 8 bits? Round your answer to the nearest whole number.

(2) What is the data amount of a 24-bit full-color image with a resolution of 320 × 480 pixels in kilobytes? Round your answer to the nearest whole number.

(3) What is the data amount of a 24-bit full-color image with a resolution of 1,920 × 1,080 pixels in megabytes? Round your answer to two decimal places.

6-9 Digital Representation and Compression Technology for Videos

Point!

1 Mechanism of Videos

(1) ([1]**Video**) : An electronic medium that creates the illusion of movement by displaying a series of still images in succession. It utilizes the ([2]**afterimage phenomenon**), which occurs due to the characteristics of human vision.

(2) ([3]**Frame**) : Each still image that composes a video.

(3) ([4]**Frame rate**) : The number of frames displayed per second. The unit is ([5]**fps**).
The ([6]**higher**) the frame rate number, the smoother the video will appear, and the ([7]**larger**) the data size will become.

(4) How to Calculate the Amount of Video Data
Amount of video data = ([8]**Amount of image data** [**B**] × **Frame rate**[**fps**] × **Time** [**seconds**])

2 Data Compression

(1) ([9]**Compression**) : A process of reducing the amount of data as much as possible while preserving the content of the data.

(2) ([10]**Decompression**) : A process of restoring compressed data to its original state.

(3) ([11]**Compression ratio**) : The extent to which data has been compressed.

$$\text{Compression ratio [\%]} = \frac{([12]\textbf{Amount of data after compression})}{([13]\textbf{Original amount of data})} \times 100$$

(4) Lossless Compression and Lossy Compression

[1] ([14]**Lossless compression**) : A compression method that allows for the complete restoration of the original data from the compressed data. Used for compressing information such as text or program data.

[2] ([15]**Lossy compression**) : A compression method in which the original data cannot be restored from the compressed data. Lossy compression is used for compressing audio, images, and video files in a way that humans do not perceive a significant difference, even if those files cannot be perfectly restored to their original form.

3 Main Types of Lossless Compression

(1) ([16]**Run-length encoding**) : A compression method that replaces sequences of the same consecutive symbols with a numerical value indicating the length of the sequence. It is effective when there are many consecutive identical symbols.

<Example> Before compression: "AAAAABBAAAABBBBBBBBAAAAAA" (25 characters)

↓

AAAAA (5 of "A") BB (2 of "B") AAAA (4 of "A") BBBBBBBB (8 of "B") AAAAAA (6 of "A")

After compression: ([17]**A5B2A4B8A6**) (10 characters)

The compression ratio is ([18] $\frac{10}{25} \times 100$) = ([19]**40 [%]**).

(2) ([20]**Huffman coding**) : A compression method that assigns shorter bit sequences to characters with higher frequencies of occurrence, and longer bit sequences to those with lower frequencies.

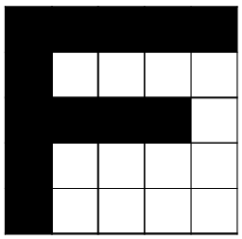
Warm Up

Answer the following questions.

(1) [1] What is the data size, in megabytes, of a 10-second uncompressed video at 30 fps, where each frame is a 24-bit full-color still image with a resolution of 500 × 200 pixels? For this question, assume that 1 KB = 1,000 B and 1 MB = 1,000 KB.

[2] Video [1] was converted into a compressed video file using a certain compression method, resulting in a file size of 30 MB. What is the compression ratio in this conversion? Round your answer to the nearest whole number and provide the answer as an integer.

(2) Consider how to represent an image using pixels. In the case of the diagram on the right, the 25 pixels in the 5 × 5 grid are read one by one from the top-left, row by row, moving to the right. If a pixel is black, write "B"; if it is white, write "W". This results in the sequence:
"BBBBBBWWWWBBBBWBWWWWBWWWW".
Next, by representing consecutive characters "B" or "W" of length n (where n is an integer of 2 or more) as "Bn" or "Wn", the sequence can be compressed to "B6W4B4WBW4BW4". Using the same method with another 5 × 5 grid, what is the alphabet letter represented by the following compressed data?



"B4WBW3B5WBW4BW4"

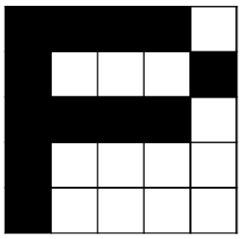
Explanation

(1) [1] The amount of data per frame is,
500 × 200 × 24 [bits] = 2,400,000 [bits]
Since 1 B = 8 bits, 2,400,000 [bits] ÷ 8 = 300,000 [B]
The amount of data of a 10-second video at 30 fps is as follows:
300,000 [B] × 30 [fps] × 10 [seconds] = 90,000,000 [B]
1 KB = 1,000 B, 1 MB = 1,000 KB, therefore,
90,000,000 [B] ÷ 1,000 ÷ 1,000 = 90 [MB] 90 MB

[2] Compression ratio [%] = $\frac{\text{Amount of data after compression}}{\text{Original amount of data}} \times 100$,

resulting in $\frac{30}{90} \times 100 = 33.3\cdots$ [%] 33%

(2) Expanding the pre-compressed data "B4WBW3B5WBW4BW4" yields "BBBBWBWWWBBBBBWBWWWWBWWWW".
If you fill in a 5 × 5 grid one pixel at a time from the top-left, row by row to the right, coloring black for "B" and white for "W", you will get the image shown on the right. Therefore, P



Try

1 Answer the following questions. For these questions, assume that 1 KB = 1,000 B and 1 MB = 1,000 KB.

(1) Complete the following sentences by filling in the blanks [1] and [5] with the appropriate terms.

> A video is a continuous display of still images. By displaying a series of gradually changing images, the phenomenon of ([1]) creates the illusion of movement for humans. The individual still images that make up a video are called ([2]), and the number of ([2]) displayed on the screen per second is referred to as the ([3]). The ([4]) the ([3]), the smoother the motion in the video appears. However, the amount of data of videos can become ([5]).

(2) What is the data size, in megabytes, of a 1-minute video at 30 fps, where each frame is a 24-bit full-color still image with a resolution of 640 × 360 pixels? Assume that the video is uncompressed. Round the amount of data to the nearest whole number and provide the answer as an integer.

2 Answer the following questions.

(1) For the following statements **A** to **D** about data compression, mark "○" if the statement is appropriate, and "×" if it is inappropriate.

A Lossless compression is a method that improves compression efficiency by allowing slight changes to the data, such as removing components that are less noticeable to humans.

B Decompression is a process of restoring compressed data to its original state.

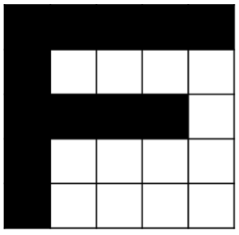
C Data compressed using lossy compression does not result in exactly the same content as the original uncompressed data.

D Data compressed using lossless compression will be exactly the same as the original data after being decompressed.

(2) When 21 MB of compressed data was decompressed, it expanded to 50 MB. What is the compression ratio of this file?

3 Answer the following questions.

(1) Consider how to represent an image using pixels. In the case of the diagram on the right, the 25 pixels in the 5 × 5 grid are read one by one from the top-left, row by row, moving to the right. If a pixel is black, write "B"; if it is white, write "W". This results in the sequence:
"BBBBBBWWWWBBBBWBWWWWBWWWW".
Next, by representing consecutive characters "B" or "W" of length n (where n is an integer of 2 or more) as "Bn" or "Wn", the sequence can be compressed to "B6W4B4WBW4BW4". Using the same method with another 5 × 5 grid, what is the alphabet letter represented by the following compressed data?
"BW3B2W3B7W3B2W3B"



(2) What is the term for the compression method that represents repeated data by listing the data and the number of consecutive occurrences, as shown in (1)? Choose one from the options **A** to **D**, and answer using the letter.

A Huffman coding **B** Codec

C Run-length encoding **D** LZ encoding

Exercise

1 Answer the following questions. For these questions, assume that 1 KB = 1,000 B and 1 MB = 1,000 KB.

(1) For the following statements **A** to **D** about video mechanisms, mark "○" if the statement is appropriate, and "×" if it is inappropriate.

A Videos create the illusion of movement by displaying a series of still images in rapid succession, leveraging the afterimage phenomenon. In reality, the images themselves are not moving.

B Each individual still image that composes a video is called a frame.

C The number of frames displayed per second is called the frame rate. The lower the frame rate, the smoother the video appears.

D When recording a video, if the resolution and frame rate settings are the same, the amount of data of a video recorded for 1 second and a video recorded for 60 seconds does not vary.

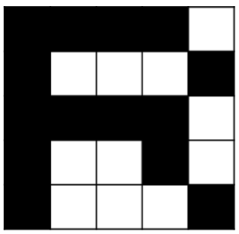
(2) What is the data size, in megabytes, of a 1-second video at 60 fps, where each frame is a 24-bit full-color still image with a resolution of 1,920 × 1,080 pixels? Assume that the video is uncompressed, and round your answer to one decimal place.

2 For the terms that fit in the blanks [1] to [3] in the following sentences, if the compression is lossless, answer with **A**; if the compression is lossy, answer with **B**.

> There are two main methods for compressing data. A compression method that allows the original data to be completely restored from the compressed data when decompressed is called ([1]). A compression method that does not allow for the complete restoration of the original data is called ([2]). For data such as images and videos, where the compressed data does not need to be perfectly restored and humans do not perceive a significant difference, ([3]) is used.

3 Answer the following questions.

(1) Consider how to represent an image using pixels. In the case of the diagram on the right, the 25 pixels in the 5 × 5 grid are read one by one from the top-left, row by row, moving to the right. If a pixel is black, write "B"; if it is white, write "W". This results in the sequence:
"BBBBWBWWWBBBBBWBWWBWBWWWB".
When "B" or "W" appears consecutively n times (where n is an integer greater than or equal to 2), they can be written as "Bn" or "Wn". Using this method, the sequence can be compressed as "B4WBW3B5WBW2BWBW3B". Using the same method with another 5 × 5 grid, what is the alphabet letter represented by the following compressed data?
"BW3BWBWBW3BW4BW4BW2"



(2) What is the term for the compression method that represents repeated data by listing the data and the number of consecutive occurrences, as shown in (1)?

6-10 Information Design

Point!

1 Information Design

(1) ([1]**Information design**) : The process of organizing and creatively expressing information according to its purpose, thereby ensuring that the intended message is correctly conveyed to the target audience.

(2) Information Design Methods

[1] ([2]**Abstraction**) : Conveying the intended information simply from within a large amount of data.

- ([3]**Pictogram**) : A pictorial symbol designed for the purpose of conveying information without using words. It is used for information signs at stations and airports.
- ([4]**Icon**) : A representation of a program's content on a computer or smartphone depicted through images or illustrations.



Pictogram



Icon

[2] ([5]**Visualization**) : Visually representing information to make it more understandable.

<Examples> Tables, graphs, etc.

[3] ([6]**Structuring**) : Organizing and expressing information clearly by arranging it according to relationships, connections, levels, stages, order, etc.

<Examples> Hierarchical menus on web pages, department store floor maps, etc.

2 Design Techniques for Ease of Understanding

(1) ([7]**User interface (UI)**) : A system for transferring information between humans and devices

<Examples> Voice input, touch panel, keyboard, etc.

[1] ([8]**CUI**) : A user interface where a device is operated by entering commands via a keyboard.

[2] ([9]**GUI**) : A user interface that allows users to issue commands in an intuitive and easy-to-understand way using icons and buttons.

(2) ([10]**User experience (UX)**) : The experience or emotional impact that users gain from interacting with a product or service.

(3) ([11]**Affordance**) : Actions or operations that can be performed on an object.

(4) ([12]**Signifier**) : Cues that prompt users to take action.

(5) ([13]**Usability**) : A measure used to indicate how easy and understandable it is for users.

(6) ([14]**Accessibility**) : The ease of access to information and services for a wide range of people.

(7) ([15]**Universal design**) : Design that is thoughtfully created with the intention that all people, regardless of age, language, nationality, or physical ability, can use it without difficulty.

3 Organizing and Classifying Information

([16]**LATCH method**) : A principle for organizing information and presenting it in a way that is easy for users to understand. Information is organized and classified according to the following five criteria.

- Location : Classification based on physical location.
- Alphabet : Classification based on alphabetical or syllabary order.
- Time : A classification based on the sequence of events in terms of time.
- Category : A classification based on the differences between things, used as a criterion to distinguish realms.
- Hierarchy : Classification based on quantitative changes such as size, level, etc.

Warm Up

Answer the following questions.

(1) Choose the most appropriate example of visualization as a method used in information design from the options **A** to **D** below, and answer using the letter.

A Pictograms for each classroom were created to guide people around the school building.

B A web page to introduce the school was created. The page was organized with menus for each hierarchy.

C The survey results were compiled into tables and graphs.

D Text summarizing multiple matters is presented in bullet points.

(2) Choose the term that best fits into the blanks [1] and [2] in the following sentences from the options **A** to **D** below, and answer using the letters.

> Current computers widely use ([1]), which allows users to manipulate graphically displayed information, such as icons, with a mouse or their finger. On the other hand, early computers used ([2]), which was a method for conveying commands by inputting characters from a keyboard.

A CUI **B** GUI **C** VUI **D** NUI

(3) What is the term for a design that is thoughtfully created in advance to allow all individuals, regardless of nationality, gender, age, or physical ability, to use it without difficulty? Choose one from the options **A** to **D** below, and answer using the letter.

A Usability **B** Universal design

C Signifier **D** User interface

Explanation

(1) **A** is an example of abstraction, **B** of structuring, **C** of visualization, and **D** of structuring.
Therefore, **C**

(2) [1]: **B** [2]: **A**

(3) **B**

Try

1 Answer the following questions.

(1) Choose the term that best fits into the blanks [1] to [3] in the following sentences from the options **A** to **E** below, and answer using the letters.

> Methods of information design include the following.
> ([1]): A method for discerning the overall picture and clearly organizing the relationships between elements.
> ([2]): A method for visually representing information.
> ([3]): A method for extracting only the necessary elements from a large amount of information.

A Specification **B** Abstraction **C** Visualization **D** Materialization **E** Structuring

(2) Choose the most appropriate example of abstraction as a method used in information design from the options **A** to **D** below, and answer using the letter.

A Pictograms for each classroom were created to guide people around the school building.
B A web page to introduce the school was created. The page was organized with menus for each hierarchy.
C The survey results were compiled into tables and graphs.
D Text summarizing multiple matters is presented in bullet points.

(3) Choose one term that represents a pictorial symbol that conveys information by abstracting the intended message without using text from the options **A** to **D** below, and answer using the letter.

A Usability **B** Affordance
C Signifier **D** Pictogram

2 Answer the following questions.

(1) Choose the term that best fits into the blanks [1] and [2] in the following sentences from the options **A** to **D** below, and answer using the letters.

> The experience that users gain through products or services is referred to as ([1]). The aim is not just to achieve ease of use, but also to achieve a sense of comfort. For that purpose, it is necessary to simplify the ([2]) that people use to operate devices and screens, choose colors with consideration for the fact that different people perceive colors differently, and add textual information.

A GUI **B** CUI **C** UX **D** UI

(2) Choose the term that best fits into the blanks [1] to [3] in the following sentences from the options **A** to **F** below, and answer using the letters.

> Regardless of the level of advancement for functions offered by a computer, those functions cannot be effectively applied in real life if they are difficult to use. The ease of use of such devices is referred to as ([1]), and ([1]) varies depending on the user. On the other hand, ease of use for a wide range of people, regardless of their abilities or conditions, is referred to as ([2]). The term ([3]) refers to design that enables ease of use for everyone, regardless of age, language, nationality, or physical ability.

A Accessibility **B** Usability **C** Affordance
D Universal design **E** User interface **F** Barrier-free

Exercise

1 Answer the following questions.

(1) Answer the following questions.

[1] Choose the most appropriate explanation of visualization as a method used in information design from the options **A** to **D** below, and answer using the letter.

A Extracting only the necessary elements from a large amount of information.

B Discerning the overall picture and clearly organizing the relationships between elements.

C Expressing the information you want to convey in a simple manner using diagrams or illustrations.

D Representing information visually.

[2] Choose all the appropriate examples of structure as a method used in information design from the options **A** to **D** below, and answer using the letters.

A A web page to introduce the school was created. The page was organized with menus for each hierarchy.

B The survey results were summarized into a graph to convey information in an easy-to-understand manner.

C Expressed a warning to the other party through illustrations without using text.

D A list was compiled to summarize the events of a particular day.

(2) What is the term for symbols used in public facilities like train stations and airports to visually convey attention or information to others, as shown in the diagram on the right?



2 Answer the following questions.

(1) Choose the term that best fits into the blanks [1] to [3] in the following sentences from the options **A** to **D** below, and answer using the letters.

([1]): A system for transferring information between humans and devices.
([2]): ([1]) through text. Operate everything from the keyboard.
([3]): ([1]) through visuals. Intuitive operation is possible using devices like a mouse.

A GUI **B** CUI **C** UX **D** UI

(2) Choose the most appropriate explanation of a signifier from the options **A** to **D** below, and answer using the letter.

A A measure used to indicate how easy and understandable it is for users.

B Design that is thoughtfully created with the assumption that everyone can use it without difficulty.

C A state that is easily accessible to a wide range of people.

D Cues that prompt users to take action.